

# Maharashtra Public Service Commission Assessment

|                              |                                             |
|------------------------------|---------------------------------------------|
| <b>Subject Name :</b>        | Associate Professor Automobile Engg Group A |
| <b>Duration :</b>            | 60                                          |
| <b>Number of Questions :</b> | 100                                         |
| <b>Total Marks :</b>         | 200                                         |
| <b>Display Marks:</b>        | Yes                                         |

|                                   |    |
|-----------------------------------|----|
| <b>Maximum Instruction Time :</b> | 0  |
| <b>Is Section Default? :</b>      | No |

**Question Number : 1 Question Id : 464662701 Question Type : MCQ**

**Insulated heat exchangers are examples of \_\_\_\_\_.**

**Options :**

1. Isolated system
2. Open system
3. Close system
4. Adiabatic system

Question Number : 2 Question Id : 464662702 Question Type : MCQ

A 2 KW electric resistance heater submerged in 5 kg of water is turned on and kept on for 10 minutes. During the process, 300 KJ of heat is lost from the water. The temperature rise of water, if specific heat of water is taken as 4.2 KJ/Kg-K will be

Options :

1. 0.43 °C
2. 43.13 °C
3. 57.43 °C
4. 71.43 °C

Question Number : 3 Question Id : 464662703 Question Type : MCQ

Which of the following statements correctly describes available energy ?

Options :

1. Total energy contained in a system
2. Energy that must be rejected to the surrounding
3. Energy lost only due to friction
4. Part of low-grade energy that can be converted into high-grade energy

Question Number : 4 Question Id : 464662704 Question Type : MCQ

A cylinder fitted with a piston has an initial volume of  $0.1 \text{ m}^3$  and contains nitrogen at 150 KPa,  $25^\circ\text{C}$ . The piston is moved, compressing the nitrogen until the pressure is 1 MPa and temperature is  $150^\circ\text{C}$ . During the compression process heat is transferred from the nitrogen, and work done on the nitrogen is 20 KJ. The amount of heat transferred to surrounding during this process will be

Options :

1. 3.6 KJ
2. 8.4 KJ
3. 7.2 KJ
4. 4.2 KJ

Question Number : 5 Question Id : 464662705 Question Type : MCQ

In the differential equation of flexure the quantity  $EI$  is often called as

Options :

1. Elastic inertia
2. Bending inertia
3. Geometric elastic modulus
4. Bending stiffness

Question Number : 6 Question Id : 464662706 Question Type : MCQ

Proof resilience is the maximum energy stored at \_\_\_\_\_.

Options :

1. Plastic limit
2. Limit of proportionality
3. Elastic limit
4. None of these

Question Number : 7 Question Id : 464662707 Question Type : MCQ

A circular bar of length (l) uniformly tapers from diameter ( $d_1$ ) at one end to diameter ( $d_2$ ) at the other end. If the bar is subjected to an axial tensile load (P), then its elongation is equal to \_\_\_\_\_.

Options :

1.  $\frac{Pl}{AE}$

2.  $\frac{Pl}{A_1 A_2 E}$

3.  $\frac{4Pl}{\pi E d_1 d_2}$

4.  $\frac{Pl}{4\pi E d_1 d_2}$

Question Number : 8 Question Id : 464662708 Question Type : MCQ

According to the maximum principal stress theory (Rankine's Theory), failure occurs when

Options :

1. Maximum principal stress reaches the limiting stress in simple tension
2. Maximum shear stress reaches the shear stress at elastic limit
3. Maximum strain exceeds the strain at fracture
4. Distortion energy reaches the yield point

Question Number : 9 Question Id : 464662709 Question Type : MCQ

The nature of shear stress distribution in a rectangular beam is

Options :

1. Uniform
2. Parabolic
3. Linear
4. Elliptic

Question Number : 10 Question Id : 464662710 Question Type : MCQ

A shaft rotating at N rps, transmits average torque T in kN-m. The power developed is

Options :

1. 
$$P = \frac{2\pi NT}{60} \text{ kW}$$

2. 
$$P = \frac{2\pi NT}{30} \text{ kW}$$

3. 
$$P = 2\pi NT \text{ kW}$$

4. 
$$P = \frac{2\pi NT}{120} \text{ kW}$$

Question Number : 11 Question Id : 464662711 Question Type : MCQ

Fatigue is a form of failure that occurs due to

Options :

1. Static loading only
2. Dynamic and fluctuating stresses
3. Composite loading only
4. Thermal loading only

Question Number : 12 Question Id : 464662712 Question Type : MCQ

Creep is best defined as

Options :

1. Instantaneous elastic deformation under load
2. Time-dependent and permanent deformation under constant load or stress
3. Sudden fracture due to impact loading
4. Plastic deformation at room temperature only

Question Number : 13 Question Id : 464662713 Question Type : MCQ

In powder metallurgy, why is the method of powder production considered critical to the success of the final product ?

Options :

1. Because it determines only the colour of the final sintered material
2. Because it eliminates the need for sintering and compaction
3. Because it controls only the porosity of the final product
4. Because it affects particle chemistry, structure and size distribution, which in turn influence compaction, sintering behaviour and final material properties

Question Number : 14 Question Id : 464662714 Question Type : MCQ

Cast iron contains graphite in nodular form is called as

Options :

1. White cast iron
2. Gray cast iron
3. Ductile cast iron
4. None of the above

Question Number : 15 Question Id : 464662715 Question Type : MCQ

Low-carbon steels generally contain carbon content of less than \_\_\_\_\_ and unresponsive to heat treatments intended to form martensite.

Options :

1. 0.50 wt%
2. 0.75 wt%
3. 0.25 wt%
4. 1.00 wt%

When developing composite powders, which of the following describes the essential requirements that the core material must satisfy ?

Options :

1. The core must be non-reactive with the chosen deposition system, catalytically active toward the deposited metal, and capable of remaining suspended during agitation
2. The core must be highly reactive, non-catalytic and insoluble in the solution
3. The core must dissolve partially during metal deposition to improve coating adhesion
4. The core must be metallic only, since non-metals cannot be coated effectively

Question Number : 17 Question Id : 464662717 Question Type : MCQ

The Fatigue strength of mild steel is \_\_\_\_\_ , if it is subjected to a very large number of stress reversals.

Options :

1. Equal to tensile strength
2. More than tensile strength
3. Equal to yield strength
4. Lower than yield strength

Question Number : 18 Question Id : 464662718 Question Type : MCQ

Convective mass transfer between a surface and a moving fluid occurs primarily due to which mechanism ?

Options :

1. Molecular diffusion only
2. Bulk fluid motion only
3. Radiation and diffusion
4. Combined effect of molecular diffusion and bulk fluid motion

Question Number : 19 Question Id : 464662719 Question Type : MCQ

The thermal conductivity of material is best described as which of the following ?

Options :

1. The rate of heat transfer per unit length per unit temperature difference through material
2. The ability of a material to store thermal energy per unit mass
3. Heat transfer coefficient between a solid surface and moving fluid
4. The amount of heat transferred per unit area per unit time irrespective of material properties

Question Number : 20 Question Id : 464662720 Question Type : MCQ

Which of the following statements regarding the emissivity of a surface is most accurate ?

Options :

1. Emissivity depends on surface characteristics and temperature and may change due to ageing effects such as oxidation or contamination
2. Emissivity depends only on the material and is unaffected by temperature
3. Emissivity is a constant property and independent of surface condition
4. Emissivity can be precisely calculated from theoretical models for all engineering surfaces

Question Number : 21 Question Id : 464662721 Question Type : MCQ

The Biot Number ( $Bi$ ) is defined as the ratio of which of the following ?

Options :

1. Convective heat transfer resistance within the body to conductive resistance at the surface
2. Internal energy storage to heat transfer rate
3. Radiative heat transfer resistance to convective heat transfer resistance
4. Thermal resistance within a body to the thermal resistance at its surface

Question Number : 22 Question Id : 464662722 Question Type : MCQ

According to Fick's law of diffusion, the rate of mass diffusion of a chemical species in a stagnant medium is directly proportional to which of the following ?

Options :

1. The total mass of the diffusing species
2. The local concentration gradient in the direction of diffusion
3. The pressure gradient of the medium
4. The temperature difference across the medium

Question Number : 23 Question Id : 464662723 Question Type : MCQ

For rotary mass balance system, where masses rotating in different parallel planes in dynamic balance if the resultant

Options :

1.  $\Sigma F = 0$
2.  $\Sigma M = 0$
3.  $\Sigma F = 0$  and  $\Sigma M = 0$
4.  $\Sigma F \neq 0$  and  $\Sigma M \neq 0$

Question Number : 24 Question Id : 464662724 Question Type : MCQ

Which of the following is an advantage of 'V' belt drive ?

Options :

1. Drive is more positive
2. Slip of the belt is less
3. More transmission of power
4. All of the above

Question Number : 25 Question Id : 464662725 Question Type : MCQ

For ductile materials, the most accurate failure theory used is generally \_\_\_\_\_.

Options :

1. Rankine
2. Tresca
3. Von Mises
4. Saint Venant

Question Number : 26 Question Id : 464662726 Question Type : MCQ

According to ASME, maximum permissible shear stress for shaft without allowance for keyways is approximately \_\_\_\_\_.

Options :

1. 42 MPa
2. 56 MPa
3. 84 MPa
4. 112 MPa

Question Number : 27 Question Id : 464662727 Question Type : MCQ

By considering creep of belt velocity ratio of belt drive is given by following equation, where  $\sigma_1$  and  $\sigma_2$ - stress in the belt on tight and slack side. E – Young's modulus for the material of the belt

Options :

1. 
$$\frac{N_2}{N_1} = \frac{d_1}{d_2} \times \frac{E + \sqrt{\sigma_2}}{E + \sqrt{\sigma_1}}$$

2. 
$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t} \left( 1 - \frac{S}{100} \right)$$

3. 
$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t}$$

4. None of the above

Question Number : 28 Question Id : 464662728 Question Type : MCQ

In case of dynamically coupled multi – DOF mass-spring system, which of the following statement is correct ?

Options :

1. the mass matrix has at least two off-diagonal term non-zero
2. the stiffness matrix has at least one non-zero off-diagonal term
3. the stiffness matrix has at least two non-zero off-diagonal term
4. the mass matrix has at least one off-diagonal term non-zero

Question Number : 29 Question Id : 464662729 Question Type : MCQ

When a hole of diameter 'd' is to be punched in a metal of thickness 't', then the force required to punch the hole is equal to

Options :

1.  $d * t * \text{Ultimate shear strength}$
2.  $\pi * d * t * \text{Ultimate shear strength}$
3.  $\left(\frac{\pi}{4}\right) * d^2 * \text{Ultimate shear strength}$
4.  $\left(\frac{\pi}{4}\right) * d^2 * t * \text{Ultimate shear strength}$

Question Number : 30 Question Id : 464662730 Question Type : MCQ

A spur gear of  $14\frac{1}{2}^\circ$  pressure angle and 48 involute teeth has pitch circle diameter 288 mm. It has/have

- a. Module of 6 mm
- b. Circular pitch of 16 mm
- c. Addendum of 6 mm
- d. Diametral pitch of 11 per mm

Options :

- 1. b and c
- 2. a and c
- 3. a and d
- 4. b and d

Question Number : 31 Question Id : 464662731 Question Type : MCQ

“Elliptical trammels” mechanism is inversion of

Options :

1. Simple 4-bar chain
2. Simple 6-bar chain
3. Single slider crank chain
4. Double slider crank chain

Question Number : 32 Question Id : 464662732 Question Type : MCQ

A centrode is defined as the locus of

Options :

1. the mass center of rotation of a rigid body as it moves in a plane
2. the instantaneous center of rotation of a rigid body as it moves in a plane
3. the geometric centre of rigid body as it moves in a plane
4. the CG of rigid body as it moves in a plane

Question Number : 33 Question Id : 464662733 Question Type : MCQ

Which of the following coupling used for transmission of power to different spindles of multiple drilling machine ?

Options :

1. Bushed pin flexible coupling
2. Oldham coupling
3. Universal Hooke's coupling
4. Flange coupling

Question Number : 34 Question Id : 464662734 Question Type : MCQ

A magnetic clutch is used in

Options :

1. the machine tool gear boxes
2. the industrial diesel drives
3. the motorcycles
4. the electric motor drives

**Question Number : 35 Question Id : 464662735 Question Type : MCQ**

A certain minimum number of teeth are to be kept for gear wheel

**Options :**

1. For better durability
2. For good size gear
3. For better strength
4. For avoiding interference and undercutting

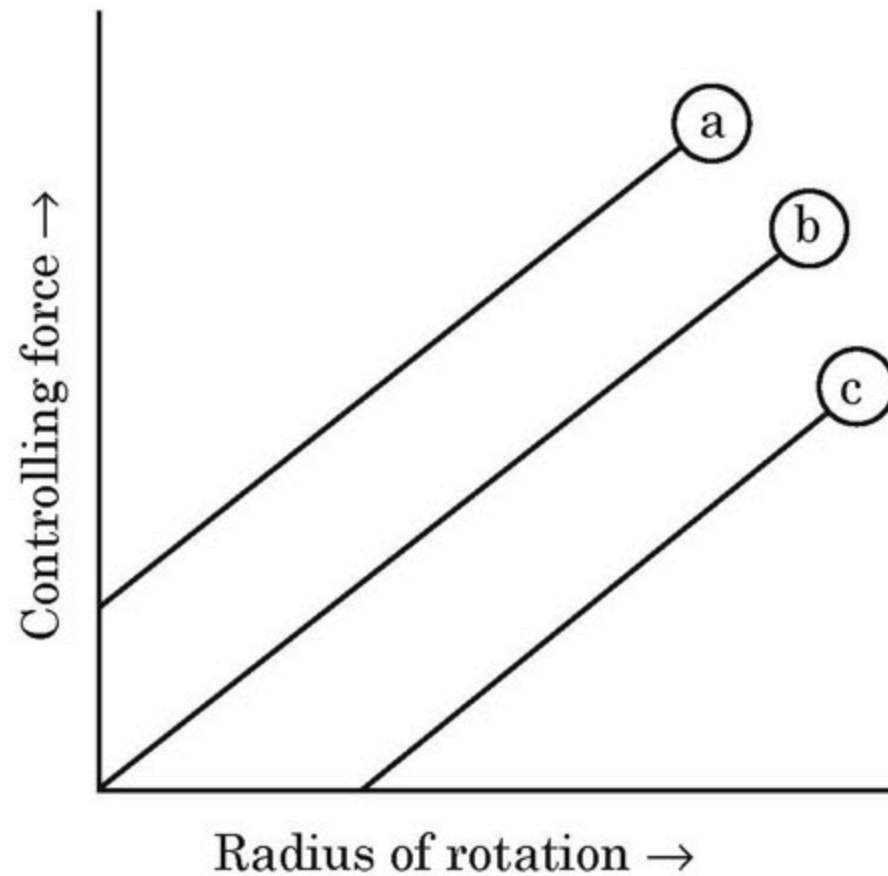
**Question Number : 36 Question Id : 464662736 Question Type : MCQ**

The swaying couple is due to

**Options :**

1. Primary unbalanced force
2. Secondary unbalanced force
3. Partial balancing
4. Two cylinders of locomotive

From the graph of Governor characteristics Line a, b and c represents



Options :

1. a – unstable, b – isochronous, c – stable
2. a – stable, b – isochronous, c – unstable
3. a – isochronous, b – stable, c – unstable

4. a – stable, b – unstable, c – isochronous

Question Number : 38 Question Id : 464662738 Question Type : MCQ

When stroke to bore ratio is less than 1, then it is called as

Options :

1. Under square
2. Over square
3. Square
4. Mini square

Question Number : 39 Question Id : 464662739 Question Type : MCQ

A gas engine working on Otto cycle has a cylinder of diameter 100 mm and stroke 125 mm. The clearance volume is 300 CC. If  $C_p = 1.004$  KJ/Kg-K and  $C_v = 0.717$  KJ/Kg-K for air, the air standard efficiency of the cycle will be

Options :

1. 44%
2. 35%
3. 28%
4. 38%

Question Number : 40 Question Id : 464662740 Question Type : MCQ

Choose the correct statement with regard to the bearing spread.

Options :

1. The distance across outside parting edges of each bearing is slightly smaller than the diameter of the housing bore
2. The diameter of each bearing is slightly greater than the diameter of housing bore
3. The distance across the outside parting edges of a bearing half is slightly greater than that of other bearing half
4. The distance across the outside parting edges of each bearing is slightly greater than the diameter of the housing bore

Question Number : 41 Question Id : 464662741 Question Type : MCQ

In dual combustion cycle the heat is added by which process ?

Options :

1. Constant volume process and constant temperature process
2. Constant pressure process and constant temperature process
3. Constant volume process and constant pressure process
4. Constant volume process and constant volume process

Question Number : 42 Question Id : 464662742 Question Type : MCQ

Which of the following characteristics is enhanced by the honing of cylinder sleeves inner surface ?

Options :

1. Cooling efficiency
2. Resistance to wear
3. Lubrication performance
4. None of these

Question Number : 43 Question Id : 464662743 Question Type : MCQ

In modern car engine the maximum oil pressure in the lubrication system is controlled by

Options :

1. Pressure switch
2. Pressure relief valve
3. Pump rotar
4. Oil filter

Question Number : 44 Question Id : 464662744 Question Type : MCQ

How many stages in the combustion process in a C.I. (Diesel) engine divided into ?

Options :

1. Two stages
2. Three stages
3. Four stages
4. Five stages

Question Number : 45 Question Id : 464662745 Question Type : MCQ

Bharat stage IV emission standard for diesel vehicle restricted Particular Matter (PM) to

Options :

1. 0.025
2. 0.015
3. 0.020
4. 0.030

Question Number : 46 Question Id : 464662746 Question Type : MCQ

If the vehicle is accelerating along the road, it is convenient to represent the effect by an equivalent inertial force known as a

Options :

1. Inertia couple
2. Centrifugal force
3. Coriolis force
4. d'Alembert force

Question Number : 47 Question Id : 464662747 Question Type : MCQ

Why does the development of vehicle-specific sensors generally take longer than that of general commercial sensors ?

Options :

1. Because vehicle sensors must operate over narrower measuring ranges

2. Because vehicle sensors are freely available and require frequent redesign and developed in parallel with the overall vehicle system

3. Because vehicle sensors use simpler designs than commercial sensors

4. Because vehicle sensors are optimized for specific applications and developed in parallel with the overall vehicle system

Question Number : 48 Question Id : 464662748 Question Type : MCQ

If vehicle speed doubles, Aerodynamic drag forces becomes

Options :

1. 2 times
2. 4 times
3. 6 times
4. 8 times

Question Number : 49 Question Id : 464662749 Question Type : MCQ

What is camber ?

Options :

1. The angle to which the steering pivot axis is tilted forward
2. The angle between vehicle's centre line and perpendicular to the rear axle
3. The angle of the wheel relative to vertical line to the road, as viewed from the front
4. None of the above

Question Number : 50 Question Id : 464662750 Question Type : MCQ

Under steady-state wind conditions, the side force imposed on a vehicle in a crosswind is given by

Options :

1.  $S_A = \frac{1}{2} \rho V^2 C_s A$

2.  $S_A = \frac{1}{2} \rho^2 V^2 C_s A$

3.  $S_A = \frac{1}{2} \rho V C_s A$

4. None of the above

Question Number : 51 Question Id : 464662751 Question Type : MCQ

In the fundamental approach to vehicle dynamics modeling, which factors primarily determine the dynamic behaviour of a vehicle during maneuvers such as acceleration, braking, ride and turning ?

Options :

1. Engine power and transmission efficiency only
2. Forces generated by tires, gravity and aerodynamics and the vehicle's response to these forces
3. Driver input alone, independent of vehicle properties
4. Road geometry without considering vehicle characteristics

Question Number : 52 Question Id : 464662752 Question Type : MCQ

Which of the following statements correctly describes the effect of tire cornering stiffness on vehicle handling characteristics ?

Options :

1. Higher front tire cornering stiffness always results in understeer and improved high-quality stability
2. Equal front and rear cornering stiffness produces neutral steer at all speeds
3. Higher rear tire cornering stiffness relative to the front is required to achieve understeer
4. Oversteer occurs only due to vehicle speed and is independent of cornering stiffness

Question Number : 53 Question Id : 464662753 Question Type : MCQ

In case of flow over a flat plate the growth of boundary layer  $\delta/x$ ,

Options :

1. decreases with an increase in the kinematic viscosity
2. increases with an increase in free stream velocity
3. decreases with an increase in the free stream velocity only if the boundary layer is laminar
4. increases with an increase in the kinematic viscosity in both laminar and turbulent boundary layer

Question Number : 54 Question Id : 464662754 Question Type : MCQ

The diameters of an impeller of centrifugal pump at inlet and outlet are 30 cm and 60 cm respectively. The minimum starting speed of pump, if it works against a head of 30 m will be

Options :

1. 1050 r.p.m.
2. 631 r.p.m.
3. 1440 r.p.m.
4. 892 r.p.m.

Question Number : 55 Question Id : 464662755 Question Type : MCQ

The flow rate through Kaplan turbine is given by \_\_\_\_\_.

Options :

1. 
$$Q = \frac{\pi}{4}(D_t^2 - D_h^2)V_F$$

2. 
$$Q = \frac{\pi}{4}(D_t^3 - D_h^3)V_F$$

3. 
$$Q = \frac{\pi}{4}d^2 \times \sqrt{2gH}$$

4. 
$$Q = \pi DB V_F$$

Question Number : 56 Question Id : 464662756 Question Type : MCQ

Which of the following statements correctly describes the characteristics and operation of a Kaplan turbine ?

Options :

1. It is suitable for high head and low specific speed applications and has many runner blades
2. It is an axial flow turbine used for low head and high specific speed, with guide blades regulating flow without changing head
3. It operates with radial flow and the governor controls the runner blade angles directly
4. It has more runner blades than a Francis turbine and is used where head is greater than 100 m

Question Number : 57 Question Id : 464662757 Question Type : MCQ

\_\_\_\_\_ is the closeness of agreement between the average value obtained from a large series of test results and an accepted reference value.

Options :

1. Trueness
2. Traceability
3. Testing
4. Stability

Question Number : 58 Question Id : 464662758 Question Type : MCQ

The fit which is obtained by making the lower limit on the hole equal to or larger than the upper limit on the shaft is known as \_\_\_\_\_.

Options :

1. Transition fit
2. Interference fit
3. Clearance fit
4. None of the above

Question Number : 59 Question Id : 464662759 Question Type : MCQ

Which statement is correct in relation with unbounded strain gauge ?

Options :

1. Freedom from faulty insulations
2. Creep and hysteresis is small (due to positive anchorage at the insulating pins)
3. Owing to absence of any organic substance, these gauges can be used in high temperature installation
4. All of the above

Question Number : 60 Question Id : 464662760 Question Type : MCQ

The Device used for measuring high temperature is known as

Options :

1. Pyrometers
2. Thermometer
3. Sling psychrometer
4. None of the above

Question Number : 61 Question Id : 464662761 Question Type : MCQ

In homogeneous co-ordinate transformation a 3D translation matrix is represented by

Options :

1.  $3 \times 4$  matrix
2.  $2 \times 2$  matrix
3.  $4 \times 4$  matrix
4.  $3 \times 3$  matrix

Question Number : 62 Question Id : 464662762 Question Type : MCQ

Translation operation in CAD refers to

Options :

1. Changing the size of an object
2. Rotating an object about an axis
3. Moving an object from one position to another without changing its orientation
4. Reflecting an object about a line

Question Number : 63 Question Id : 464662763 Question Type : MCQ

In NC machine, the motion control system is

Options :

1. Point to Point type
2. Straight cut type
3. Contouring type
4. All of the above

Question Number : 64 Question Id : 464662764 Question Type : MCQ

In case of 3D-modelling, once a geometric model has been designed it can be used as a component in more complex models. This is referred to as

Options :

1. Sequential modelling
2. Parametric modelling
3. Hierarchical modelling
4. Assembly modelling

Question Number : 65 Question Id : 464662765 Question Type : MCQ

\_\_\_\_\_ is the international standard for attachment point for child safety seats in passenger cars.

Options :

1. ISOFIX
2. ISOCIX
3. ISOPIX
4. ISOXXX

Question Number : 66 Question Id : 464662766 Question Type : MCQ

Which one of the following is the advantages of single plate clutch ?

Options :

1. As compare to cone clutch, the springs have to be more stiff
2. Greater force required to be applied by the driver while disengaging
3. It is not reliable
4. Gear changing is easier than cone clutch

Question Number : 67 Question Id : 464662767 Question Type : MCQ

Which of the following is the advantages of rubber springs suspension ?

Options :

1. Excellent vibration damping properties
2. Rubber is more reliable
3. It can store greater energy per unit weight than the steel
4. All of the above

Question Number : 68 Question Id : 464662768 Question Type : MCQ

Which of the following suspension gives maximum room in the engine compartment ?

Options :

1. Mac Pherson strut type suspension
2. Wishbone type or parallel link type suspension
3. Vertical guide type suspension
4. Trailing link type suspension

Question Number : 69 Question Id : 464662769 Question Type : MCQ

The difference between DOT 3 and DOT 4 fluid is that the \_\_\_\_\_.

Options :

1. DOT 4 fluid has a higher boiling point than DOT 3 fluid
2. DOT 3 fluid has a higher boiling point than DOT 4 fluid
3. DOT 3 and DOT 4 have same boiling point, but DOT 4 fluid has a longer service life
4. DOT 4 fluid is more resistant to freezing than DOT 3 fluid

Question Number : 70 Question Id : 464662770 Question Type : MCQ

Which of the following axle transmit driving torque only and load of vehicle with stand by wheel and axle casing ?

Options :

1. Semi-floating axle
2. Three-quarter floating axle
3. Full-floating axle
4. None of the above

Question Number : 71 Question Id : 464662771 Question Type : MCQ

When the rear wheels are jacked up and gears are in neutral, turning one rear wheel forward in a rear drive vehicle will cause the other wheel to

Options :

1. Turn backward
2. Turn forward
3. Turn in same direction
4. Remain stationary

**Question Number : 72 Question Id : 464662772 Question Type : MCQ**

If the interpolation function chosen for the co-ordinates is of a lower order than the one for the displacement then the elements are

**Options :**

1. Superparametric
2. Subparametric
3. Isoparametric
4. All of the above

**Question Number : 73 Question Id : 464662773 Question Type : MCQ**

To find the nodal displacements in all parts of the element, \_\_\_\_\_ is used.

**Options :**

1. Shape function
2. Node function
3. Element function
4. Co-ordinate function

Question Number : 74 Question Id : 464662774 Question Type : MCQ

In 2D finite element analysis, the basic element used for plane stress/strain problems is \_\_\_\_\_.

Options :

1. Line elements
2. Beam elements
3. Axisymmetric elements
4. Triangular or quadrilateral elements

Question Number : 75 Question Id : 464662775 Question Type : MCQ

Shape functions in FEM are used to

Options :

1. Apply boundary conditions directly
2. Define material properties
3. Interpolate field variables at interior points from nodal values
4. Compute stresses only

**Question Number : 76 Question Id : 464662776 Question Type : MCQ**

In finite element analysis, the Jacobian matrix used to

**Options :**

1. Transformation of co-ordinates from local to global system
2. Transformation of co-ordinates from global to local system
3. Transformation of derivatives from global to local system
4. Transformation of derivatives from local to global system

**Question Number : 77 Question Id : 464662777 Question Type : MCQ**

The art of subdividing the structure into a convenient number of smaller elements is known as

**Options :**

1. Assemblage
2. Continuum
3. Traction
4. Discretization

Question Number : 78 Question Id : 464662778 Question Type : MCQ

A truss is a structure that is made of straight structural members capable of carrying loads only in their

Options :

1. Own direction
2. Own direction and shear direction
3. Shear direction
4. Shear directions and moments around their axes

Question Number : 79 Question Id : 464662779 Question Type : MCQ

Which of the following statements is/are correct relating to Spongy Brake ?

- a. Air in hydraulic system
- b. Brake shoes out of adjustment
- c. Defective master cylinder
- d. Loose connections, damaged brake line and loss of brake fluid

Options :

- 1. Only a and c
- 2. Only b, c and d
- 3. Only a, b and c
- 4. All of the above

Question Number : 80 Question Id : 464662780 Question Type : MCQ

Conformity of Production (CoP) ensures that \_\_\_\_\_.

Options :

1. Every vehicle model undergoes crash testing
2. Only prototype vehicles meet standards
3. Mass produced vehicles continue to comply with the approved type
4. Vehicles are approved only once during design stage

Question Number : 81 Question Id : 464662781 Question Type : MCQ

The method used for 24 hours SO<sub>2</sub> measurement prescribed by CPCB is

Options :

1. Gravimetric
2. Ultraviolet Fluorescence
3. Chemiluminescence
4. Indophenol blue

Question Number : 82 Question Id : 464662782 Question Type : MCQ

Homologation categories in India includes \_\_\_\_\_.

Options :

1. Category A (Agriculture tractors), Category C (Construction equipment vehicles), Category L (Motor cycles)
2. Category X (Export vehicles), Category Y (Luxury vehicles)
3. Category P (Passenger cars), Category B (Buses)
4. Category S (Sports cars), Category E (Electric vehicles only)

Question Number : 83 Question Id : 464662783 Question Type : MCQ

According to AIS, Category M1 vehicles are defined as

Options :

1. Goods vehicle  $\leq 3.5$  t
2. Passenger vehicles  $\leq 8$  seats + driver
3. Passenger vehicles  $> 8$  seats  $\leq 5$  t
4. Passenger vehicles  $> 8$  seats  $> 5$  t

The correct way to measure a Camshaft's total runout dimension is to \_\_\_\_\_.

Options :

1. Rotate Camshaft on a cylinder head plate and measure total runout dimension using a dial indicator
2. Support the journal at each end of Camshaft in v-block, then rotate Camshaft and measure total runout dimension using a dial indicator
3. Support journal at each end of the Camshaft in a v-block, then rotate Camshaft and measure total runout dimension using a micrometer
4. Support cam lobe at each end of Camshaft in v-block, then rotate Camshaft and measure the total runout dimension using a micrometer

Question Number : 85 Question Id : 464662785 Question Type : MCQ

Which of the following is most likely cause of build-up of dry, black ash or soot in exhaust pipe ?

Options :

1. An overly rich air-fuel mixture
2. An overly lean air-fuel mixture
3. Oil consumption
4. Exhaust pipe tagging

Question Number : 86 Question Id : 464662786 Question Type : MCQ

Purpose of Assistance of Automobile Industry Standards Committee (AISC) to Central Motor Vehicle Rule (CMVR) is \_\_\_\_\_.

Options :

1. To prepare new standards for automotive items related to safety
2. To prepare standards for electronic items
3. To prepare standards for aeronautical components
4. To prepare standards for components of mass production

Question Number : 87 Question Id : 464662787 Question Type : MCQ

Which three-terminal power semiconductor device primarily used as a fast-acting, high-efficiency electronic switch in motors and control systems of EV ?

Options :

1. SCR
2. IGBT
3. BJT
4. MOSFET

Question Number : 88 Question Id : 464662788 Question Type : MCQ

In parallel hybrid systems it is useful to define a variable called as 'Degree of Hybridisation' (DOH) and it is given by an equation \_\_\_\_\_.

Options :

1. 
$$\text{DOH} = \frac{\text{Electric motor power}}{\text{Electric motor power} + \text{IC engine power}}$$

2. 
$$\text{DOH} = \frac{\text{Electric motor power}}{\text{IC engine power}}$$

3. 
$$\text{DOH} = \frac{\text{IC engine power}}{\text{Electric motor power}}$$

4. None of the above

Question Number : 89 Question Id : 464662789 Question Type : MCQ

A phrase “Range anxiety” often used in electric vehicles refers to \_\_\_\_\_.

Options :

1. Fear about high vehicle speed
2. Fear about battery explosion
3. Fear about the distance an EV can drive and the worry that it may not be enough to reach out destination
4. None of these

Question Number : 90 Question Id : 464662790 Question Type : MCQ

The battery management controller helps prevent battery damage by protecting the cell against \_\_\_\_\_.

Options :

1. Increasing charging current
2. Overcharging and deep discharge
3. Increasing charging voltage
4. None of the above

Question Number : 91 Question Id : 464662791 Question Type : MCQ

Which emerging control technique is used to enhance torque response and improve efficiency in electric vehicle drives ?

Options :

1. Close-loop control
2. Automatic control
3. Scalar control
4. Field-oriented control or vector control

Question Number : 92 Question Id : 464662792 Question Type : MCQ

Which of the following fuel cell having maximum operating temperature range when solid electrolyte is used ?

Options :

1. PEMF (Polymer Exchange Membrane Fuel Cells)
2. MCFC (Molten Carbonate Fuel Cells)
3. SOFC (Solid Oxide Fuel Cells)
4. DMFC (Direct Methanol Fuel Cells)

Question Number : 93 Question Id : 464662793 Question Type : MCQ

Storing hydrogen on board a vehicle is to liquefy the gas at cryogenic temperature of \_\_\_\_\_ is technically challenging.

Options :

1.  $-150^{\circ}\text{C}$
2.  $-190^{\circ}\text{C}$
3.  $-259.2^{\circ}\text{C}$
4.  $-179.2^{\circ}\text{C}$

Question Number : 94 Question Id : 464662794 Question Type : MCQ

Which is the most widely used electric motor type in a typical modern passenger car for applications like high performance and operable in a wide speed range ?

Options :

1. PMSM (Permanent Magnet Synchronous Motor)
2. DC Induction Motor
3. BLDC (Brushless DC Motor)
4. Stepper Motor

Question Number : 95 Question Id : 464662795 Question Type : MCQ

Each stop lamp on a passenger car and on multipurpose passenger vehicles, truck, trailer or bus shall have an effective luminous area not less than \_\_\_\_\_ in the intended direction.

Options :

1. 22.5 sq.cms
2. 21.5 sq.cms
3. 22.00 sq.cms
4. 21.00 sq.cms

Question Number : 96 Question Id : 464662796 Question Type : MCQ

The continuous improvement is focussed in 'Kaizen' systems and the essence of 'Kaizen' lies at

Options :

1. Small projects that seek to build success and confidence
2. Elephant size projects which loses confidence
3. Eliminating all employees involvement
4. None of the above

Question Number : 97 Question Id : 464662797 Question Type : MCQ

During vehicle accident, approximately at what time after impact, the air bag is fully inflated and driver's momentum will be absorbed by the airbag ?

Options :

1. 15 ms
2. 30 ms
3. 40 ms
4. 120 ms

Question Number : 98 Question Id : 464662798 Question Type : MCQ

The purpose of Anti-lock Braking System (ABS) in vehicle is

Options :

1. To make the vehicle accelerate faster
2. To prevent the vehicle wheels from locking when the brakes are applied
3. To improve fuel efficiency
4. To improve comfort of the ride

Question Number : 99 Question Id : 464662799 Question Type : MCQ

The wiper system on a vehicle shall be capable of attaining the minimum operating frequency of \_\_\_\_\_ regardless of engine load and engine speed.

Options :

1. 5 cycles/min.
2. 10 cycles/min.
3. 15 cycles/min.
4. 20 cycles/min.

Question Number : 100 Question Id : 464662800 Question Type : MCQ

As per the test procedures mentioned in Sl. No. 10 of clause no. 4, the photometric and colorimetric measurements for regulatory compliance may be made on how many samples ?

Options :

1. Three samples
2. Five samples
3. Seven samples
4. Nine samples